

Jake Ward

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Research Experience

Anthropic Fellows Program

August 2026–Present

Fellow [Shawn Wang & Adam Jermy]

- Studying how loss-masking can affect generalization in character training.

MATS

January 2026–August 2026

Fellow [Alex Cloud & Alex Turner]

- Developed GRAFT, an alignment method which leverages weak monitors to localize and prevent reward hacking during RL [1].

MATS

June 2025–December 2025

Fellow [Adam Shai & Paul Riechers]

- Developed novel methods for interpreting reasoning models using minimal parameter adapters [4].
- Studied how LLMs represent distributions over their beliefs about the world.

AISC

February 2025–April 2025

Mentee [Kola Ayonrinde (UK AISI)]

- Explored whether SAEs represent entity-attribute binding vectors.
- Examined dynamics of SAE feature splitting using semi-synthetic data.

University of Rhode Island

March 2019–July 2019

Research Assistant [Marco Alvarez]

- Developed and optimized ML pipelines for subject age estimation.

Publications

- [1] **Jake Ward***, Shawn Hu*, Aria Wong, Nathan Hu, Bryce Woodworth, Alexander Matt Turner, Alex Cloud. “Gradient Routing Localizes and Removes Unintended Behaviors in RL.” *NeurIPS*, 2026. (Poster)
- [2] Satchel Grant, Victor Gillioz, **Jake Ward**, Thomas McGrath. “Shifting the Gradient: Understanding How Defensive Training Methods Protect Language Model Integrity.” *NeurIPS*, 2026. (Poster)
- [3] Nathan Hu, **Jake Ward**, Thomas Icard, Christopher Potts. “Transcoder Adapters for Reasoning-Model Diffing.” *NeurIPS*, 2026. (Poster)
- [4] **Jake Ward**, Paul Riechers, Adam Shai. “Rank-1 LoRAs Encode Interpretable Reasoning Signals.” *NeurIPS Workshop on Mechanistic Interpretability*, 2025.
- [5] **Jake Ward**. “Reasoning Elicitation is Scale-Dependent.” *NeurIPS Workshop on Efficient Reasoning*, 2025.
- [6] **Jake Ward***, Chuqiao Lin*, Constantin Venhoff, Neel Nanda. “Reasoning-Finetuning Repurposes Latent Representations in Base Models.” *ICML Workshop on Actionable Interpretability*, 2025.

Industry Experience

Melange

February 2021–September 2023

Co-founder & Lead Engineer

- Co-founded a venture-backed startup (\$10M seed) building an auction-based prediction markets platform.
- Later pivoted to building LLM agents for patent prosecution.
- Led our engineering team of ~5 engineers.

Spark Insights

July 2019–May 2021

ML Engineer

- Developed computer vision models to identify and predict damage to properties after natural disasters.

Education

University of Rhode Island

2015–2019

B.S. Computer Science